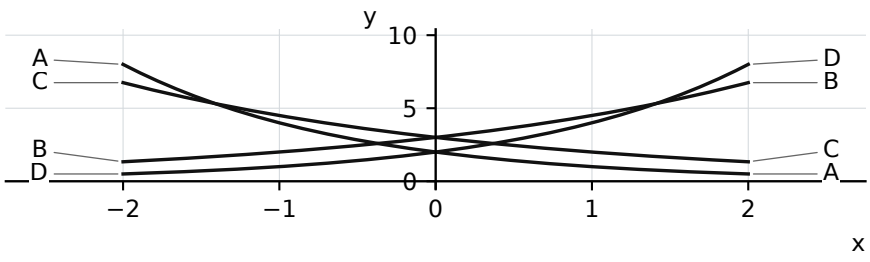
MATH 101 Unit 4: Exponentials, Logarithms, and Systems

Practice 2 | Version 202610122020.44 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

QUESTIONS	ANSWERS
1. Solve both equations exactly. Check the logarithmic domain in (b). $(a)\quad 4^{x-2} = 2^{x+3}$ $(b)\quad \log_3(x-2) + \log_3(4) = 2$	1. (a) 8 pts (b), with domain 12 pts
2. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10. $(a)\quad \log(0.4) + \log(2500)$ $(b)\quad \frac{\log_4(27)}{\log_4(9)}$	2. (a) 10 pts (b) 10 pts
3. (a) Solve the linear system. (b) Solve the nonlinear system algebraically. (c) Test whether (0,0) belongs to the region given by the inequalities; show both comparisons. $(a)\quad x + y = 5$ $x - y = 1$ $(b)\quad y = x^2$ $y = x + 2$ $(c)\quad y \geq x^2,\ y < x + 2$	3. (a) Ordered pair 6 pts (b) Intersections 10 pts (c) Membership 4 pts
4. Match each function to graph A, B, C, or D. Each letter is used once. $(a)\quad 3(\frac{2}{3})^x$ $(b)\quad 2(\frac{1}{2})^x$ $(c)\quad 2(2)^x$ $(d)\quad 3(\frac{3}{2})^x$ 	4. (a) 5 pts (b) 5 pts (c) 5 pts (d) 5 pts

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QUESTIONS	ANSWERS
5. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain. $f(x) = \log_3(x - 2) + 2$	5. (a) Domain 4 pts (a) Range 4 pts (b) Asymptote 3 pts (b) x-intercept 3 pts (c) Left end 3 pts (c) Right end 3 pts

Instructor answer key: MATH 101 Unit 4: Exponentials, Logarithms, and Systems

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a)

$x = 7$

8 pts

(b), with domain

$x = \frac{17}{4};\ x > 2$

12 pts

2. (a)

3

10 pts

(b)

$\frac{3}{2}$

10 pts

3. (a) Ordered pair

$(3,\ 2)$

6 pts

(b) Intersections

$(-1,\ 1);\ (2,\ 4)$

10 pts

(c) Membership

At (0,0): 0 >= 0 and 0 < 2; yes, in the region.

4 pts

4. (a)

C

5 pts

(b)

A

5 pts

(c)

D

5 pts

(d)

B

5 pts

TEST PAGE 2

5. (a) Domain

$(2, \infty)$

4 pts

(a) Range

$(-\infty, \infty)$

4 pts

(b) Asymptote

$x = 2$

3 pts

(b) x-intercept

$(\frac{19}{9},\ 0)$

3 pts

(c) Left end

$x \to 2^{+}:\ f(x) \to -\infty$

3 pts

(c) Right end

$x \to \infty:\ f(x) \to \infty$

3 pts